CRIME DATA EXPLORATORY ANALYSIS REPORT

Prepared for Nadine Green

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Executive Summary

This report examines crime patterns across five selected UK police forces using street-level crime data obtained from data.police.uk. The purpose of this analysis is to support the stakeholder, Nadine Green, Head of Sales at a real estate company, in identifying locations that may be desirable for property investment and residential living.

The dataset covers the period January 2024 to December 2025 and contains records of reported crimes across the selected police force regions. The data was cleaned and combined into a single dataset before being analysed using Python. Exploratory data analysis techniques were applied to investigate differences in crime levels across regions, examine how crime changes over time, identify the most common crime types, and visualise geographic crime hotspots.

Key findings include:

- Crime levels vary noticeably across the five police forces, with some regions reporting higher overall volumes of crime.
- Crime levels fluctuate month-to-month but remain relatively stable over the two years.
- Violence and sexual offences represent the most common crime category across the dataset.
- Crime type distributions are broadly similar across regions, although the proportions of certain crime types vary slightly.
- Geographic visualisation highlights clusters of crime activity in certain areas, indicating potential hotspots.

Based on the findings of this exploratory analysis, further investigation should prioritise Surrey Police and Thames Valley Police, which show substantially lower crime volumes than the larger metropolitan regions.

Metric	Value
Total Crimes	7,289,719
Police forces analysed	5
Crime categories	14
Most common crime	Violence & sexual offences
Lowest crime force	Surrey

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2. Introduction

This project analyses crime data in the UK to provide insight into which areas may be more or less desirable to live in. The stakeholder, Nadine Green, has requested this analysis to support decision-making within a real estate company by identifying regions with relatively lower levels of crime.

The analysis focuses on the crime patterns of five selected police forces. These forces were chosen to provide a comparison across different regions and policing areas while maintaining a manageable dataset size for analysis.

The selected police forces are:
• Metropolitan Police Service
• Surrey Police
• Thames Valley Police
• West Midlands Police
• West Yorkshire Police

Crime levels vary significantly across the five police forces. The Metropolitan Police Service recorded over 2 million crimes during the study period, while Surrey Police recorded less than 200,000 incidents, representing over 90% fewer crimes. These police forces were selected to provide a comparison between different geographic and population contexts. The Metropolitan Police Service represents a large metropolitan policing area with high population density, while Surrey Police and Thames Valley Police represent areas with lower population density and suburban characteristics. West Midlands Police and West Yorkshire Police provide additional comparison from major urban regions outside London. Selecting these forces allows crime patterns to be compared across both metropolitan and regional policing environments.

This analysis focuses on the time period January 2024 - December 2025, allowing recent crime patterns to be examined over two years.

This project aims to address the following objectives:
• What crimes are the most prevalent in each region?
• Which regions report the highest levels of crime over time?
• Are there identifiable geographic hotspots where crime occurs more frequently?

3. Data Pre-Processing

Before conducting the exploratory data analysis, the raw crime datasets were cleaned and prepared to ensure the data were suitable for analysis. The raw data consisted of multiple CSV files downloaded from the UK Police open data portal. These files contained monthly street crime records and outcome records for the selected police forces.

The datasets were combined into a single dataset using Python. Several data preparation steps were then performed, including reviewing missing values, removing unnecessary columns, handling duplicate records, and ensuring variables were stored in appropriate formats.

3.1 Data loading and combining

The raw data was stored across multiple folders containing monthly CSV files for each police force. These files included both street crime records and crime outcome records.

A Python script was used to automatically iterate through the folders, load each CSV file, and combine them into a single dataset. During this process, the police force name was extracted from the file name and added as a new column so that each record could be associated with the correct police force.

After combining all files, the dataset contained 7,783,098 rows and 14 columns, representing crime incidents and outcomes recorded between January 2024 and December 2025.

3.2 Initial Data Inspection

After the datasets were combined, an initial inspection of the data was performed to understand the structure of the dataset. This included reviewing the column names, data types, and a sample of the records.

The dataset included information such as the crime identifier, reporting police force, geographic location, crime category, and the outcome of the investigation. The dataset also included geographic identifiers such as longitude, latitude, and Lower Layer Super Output Area (LSOA) codes. During this stage, missing values and duplicate records were identified for further cleaning.

3.3 Handling Missing Values

Several columns contained missing values in the raw dataset. Some missing values were expected because the dataset combines two types of records: crime incidents and outcome records. Crime records include a crime category but may not yet have an outcome, while outcome records may not include a crime category.

To ensure the dataset could be analysed effectively, missing values were handled as follows:

- Missing values in the Crime ID column were replaced with “Unknown” to preserve the record.
- Missing values in LSOA code and LSOA name were replaced with “Unknown”.
- Missing values in Last outcome category were labelled as “Outcome unknown”.
- Missing values in Outcome type were labelled as “Outcome not recorded”.

These steps ensured that records were not removed unnecessarily and that missing values were clearly identifiable during analysis.

3.4 Distinguishing Crime and Outcome Records

As the dataset combines crime records and outcome records, some fields are populated only for specific record types. For example, crime records contain values in the Crime type column, while outcome records primarily contain information in the Outcome type column.

To differentiate between these two record types, a new column called Record type was created. Records with a valid crime category were labelled as Crime**,** while records representing investigation outcomes were labelled as Outcome**.**

This distinction allows the analysis to separately examine crime incidents and investigation outcomes while keeping the dataset unified.

3.5 Removing Duplicates

After combining multiple CSV files, duplicate rows were identified within the dataset. Duplicate records may occur when data from multiple sources overlaps or when files contain repeated entries.

A duplicate check revealed 493,379 duplicate rows, which were subsequently removed from the dataset. Removing these duplicates ensured that each crime record was counted only once during the analysis. After removing duplicates, the dataset contained 7,289,719 unique records.

3.6 Final Data Validation

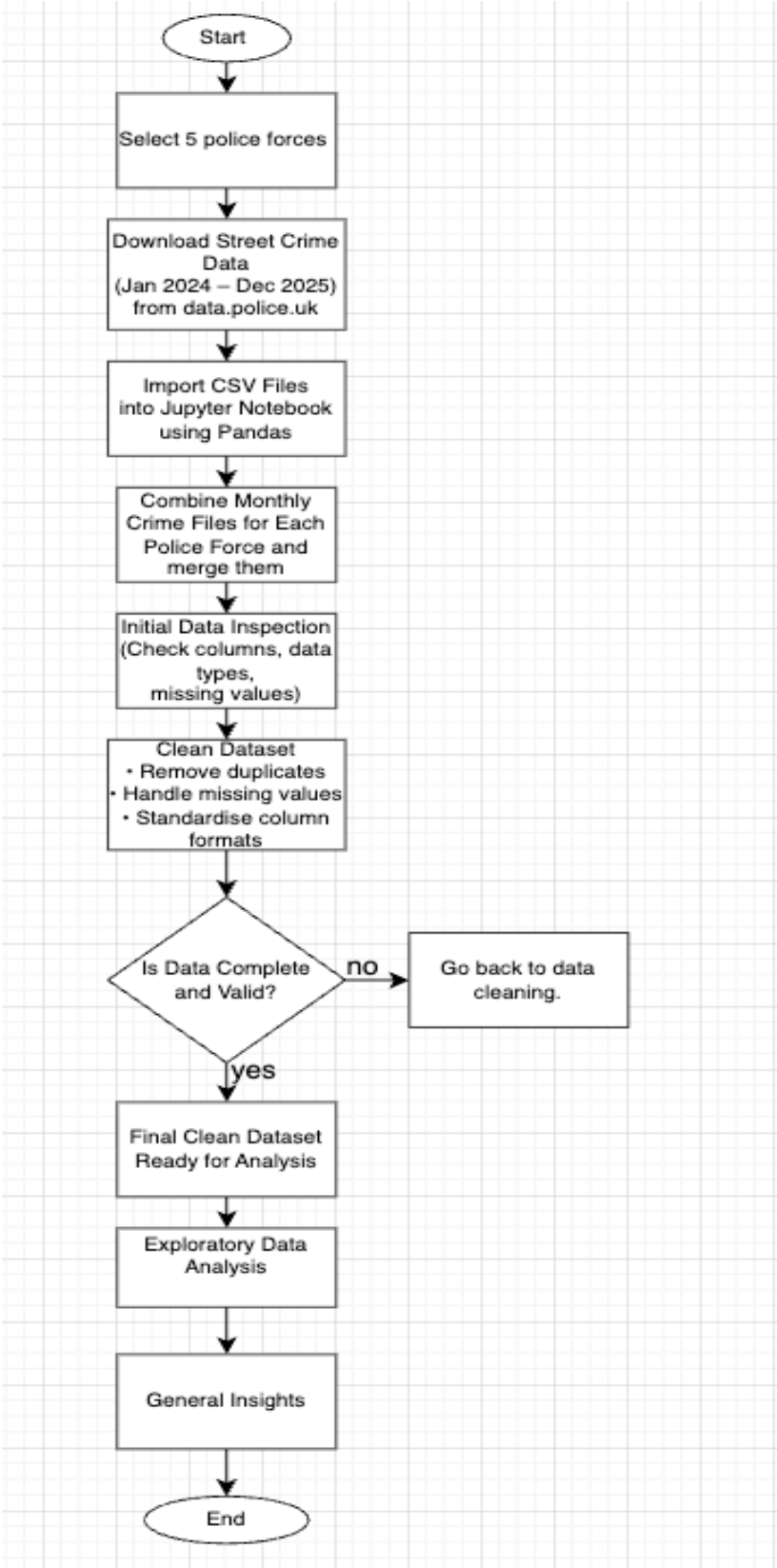
Following the data cleaning process, the dataset was reviewed again to confirm that the preprocessing steps were successful.

The final dataset contains:

- 14 variables
- 7,289,719 rows
- No duplicate records

A small number of records still contain missing longitude and latitude values. These missing values are common in police datasets due to location anonymisation and were retained because they do not affect the majority of the analysis. The cleaned dataset is now ready for exploratory data analysis.

3.7 Workflow


Start
Select 5 police forces
Download Street Crime Data (Jan 2024 – Dec 2025) from data.police.uk
Import CSV Files into Jupyter Notebook using Pandas
Combine Monthly Crime Files for Each Police Force and merge them
Initial Data Inspection (Check columns, data types, missing values)
Clean Dataset
• Remove duplicates
• Handle missing values
• Standardise column formats
Is Data Complete and Valid?
no
Go back to data cleaning.
yes
Final Clean Dataset Ready for Analysis
Exploratory Data Analysis
General Insights
End

4. Exploratory Data Analysis

To begin the exploratory analysis, the dataset was examined to identify patterns in crime levels across police forces. The analysis focuses on overall crime levels, trends, crime types, investigation outcomes and geographic distribution.

4.1 Crime Counts by Police Force

The first stage of the analysis examines the overall distribution of crimes across police forces.

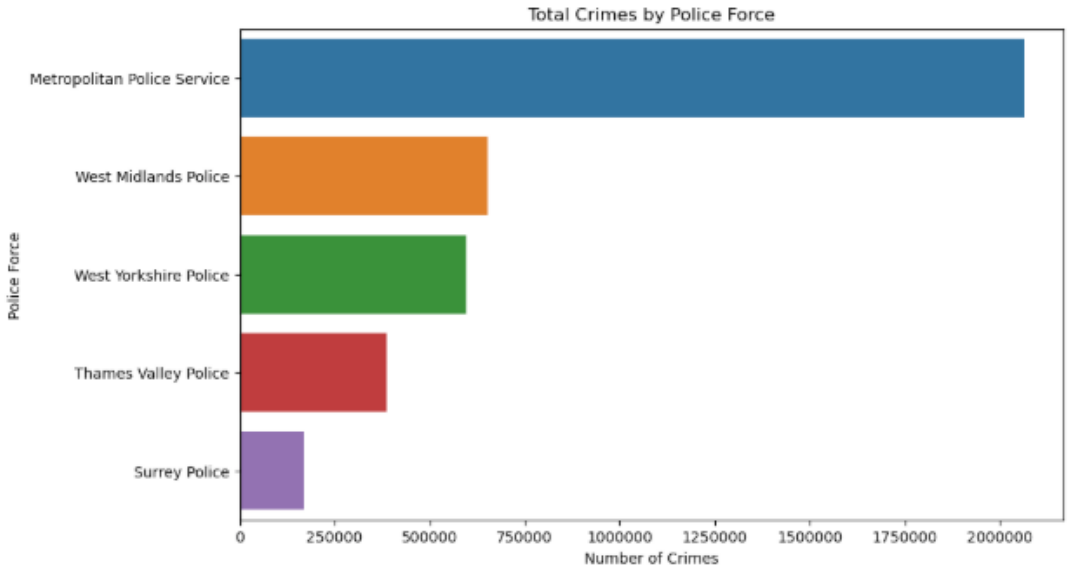


Figure 1:Total number of recorded crimes by police force (January 2024 – December 2025)

Figure 1 displays the total number of crimes recorded by each police force during the study period. The Metropolitan Police Service reports the highest number of crimes by a significant margin, which is expected given the large population and dense urban environment within London. West Midlands Police and West Yorkshire Police also report relatively high numbers of crimes, while Thames Valley Police and Surrey Police record noticeably lower totals.

These differences provide an initial indication of how crime volume varies across regions, although it is important to consider that population size and urban density can strongly influence the number of crimes reported.

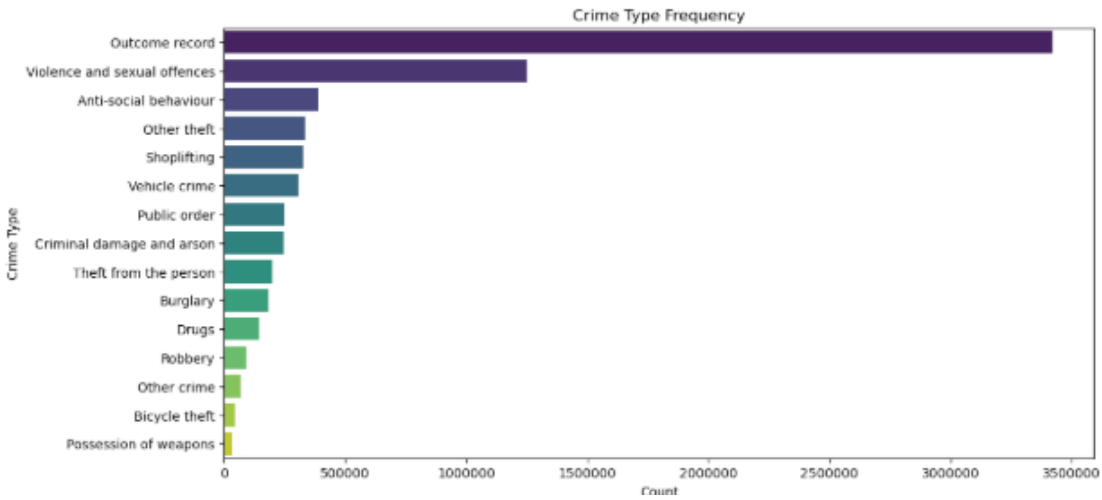


Figure 2 : Distribution of crime types recorded across the dataset

Figure 2 illustrates the frequency of different crime categories across the dataset. The most prevalent category is Violence and sexual offences, which appears substantially more frequently than any other crime type. This is followed by Anti-social behaviour, other theft, and Shoplifting. Less frequent categories include Robbery, Bicycle theft, and Possession of weapons.

Understanding the distribution of crime types provides useful context for later analysis, as it highlights which offences contribute most significantly to overall crime levels across the selected police forces.

4.2 Monthly Crime Counts by Police Force

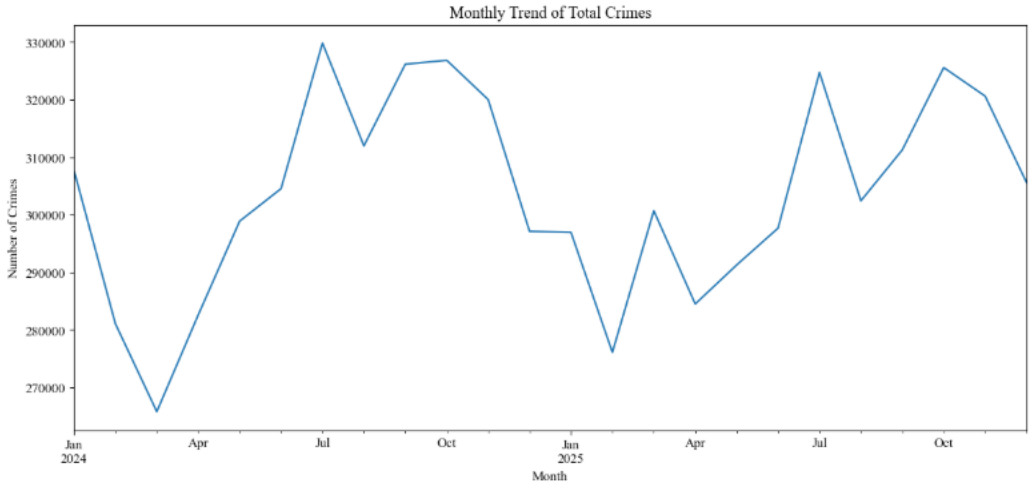


Figure 3 : Monthly trend of total recorded crimes across all selected police forces (January 2024 – December 2025)

Figure 3 displays the total number of crimes recorded each month across all selected police forces combined. The chart shows that crime levels fluctuate throughout the study period but remain broadly within a similar range. The lowest levels of crime appear early in the period, while several peaks occur during mid-2024 and mid-2025.

These fluctuations may reflect seasonal patterns or wider socio-economic factors influencing crime levels. However, the overall pattern suggests that crime levels remain relatively stable across the two-year period without any sustained upward or downward trend.

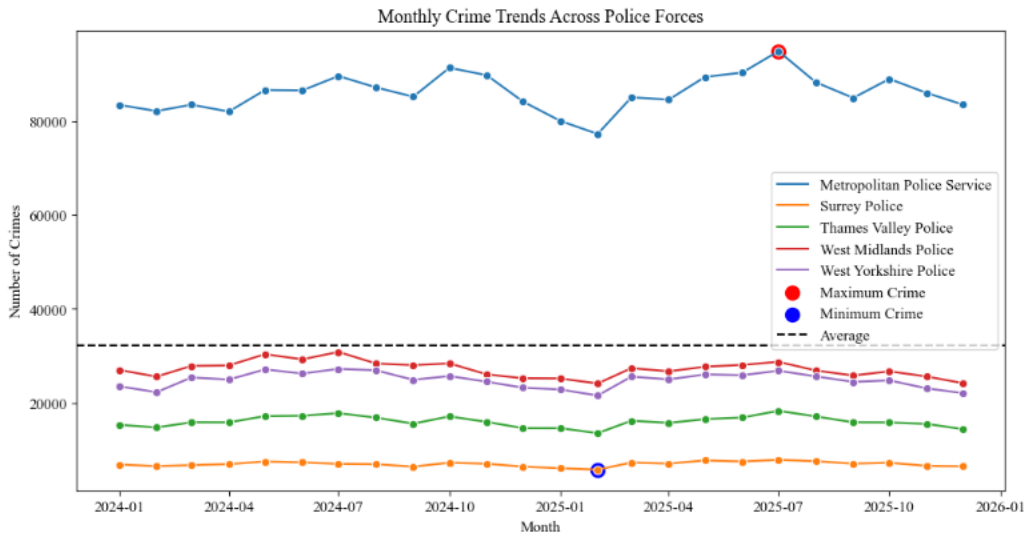


Figure 4 :Monthly crime counts by police force, highlighting maximum, minimum, and average crime levels.

Figure 4 compares the monthly crime counts for each police force over the study period. The Metropolitan Police Service consistently records the highest number of crimes each month, which is expected given the size and population density of London. In contrast, Surrey Police report the lowest number of crimes across the period.

The chart also highlights the maximum and minimum crime values observed in the dataset, as well as the average crime level across the selected police forces. Although the absolute crime counts differ between regions, the overall trends tend to rise and fall at similar times across the police forces. This suggests that broader regional influences, such as seasonal factors, may affect crime patterns across multiple areas simultaneously.

4.3 Crime type Distribution

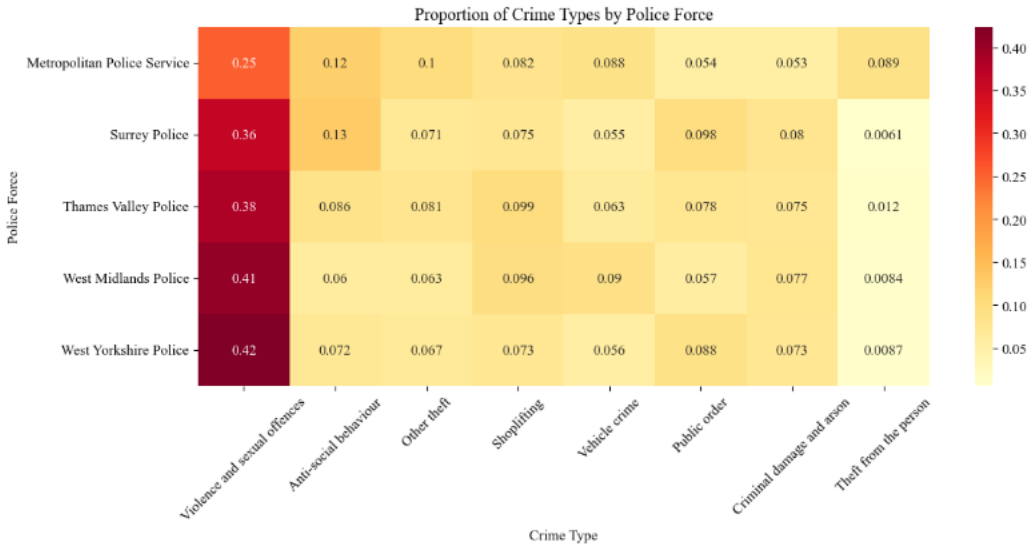


Figure 5: Proportion of crime types recorded across the five police forces.

Figure 5 presents a heatmap illustrating the proportion of different crime types recorded across the five selected police forces. Using proportions rather than raw counts allows a clearer comparison between regions, as the total number of crimes differs significantly between police forces.

Across all regions, Violence and sexual offences represent the largest proportion of recorded crimes. This category accounts for approximately 25% of crimes within the Metropolitan Police Service, while reaching over 40% of crimes within West Midlands Police and West Yorkshire Police. This suggests that violent offences form a particularly large share of recorded crime within these regions.

Other common categories include Anti-social behaviour, Other theft, Shoplifting and Vehicle crime, which appear consistently across all police forces but in slightly different proportions. For example, the Metropolitan Police Service records a higher proportion of Theft from the person, while Surrey Police and Thames Valley Police show relatively lower proportions of this offence.

Overall, while the same broad categories of crime appear across all police forces, the composition of crime varies between regions. These differences highlight how the nature of crime can differ between policing areas even when the overall types of offences remain similar. Across the full dataset, violence and sexual offences account for approximately 32% of all recorded crimes, making this the dominant crime category overall.

4.4 Crime Locations

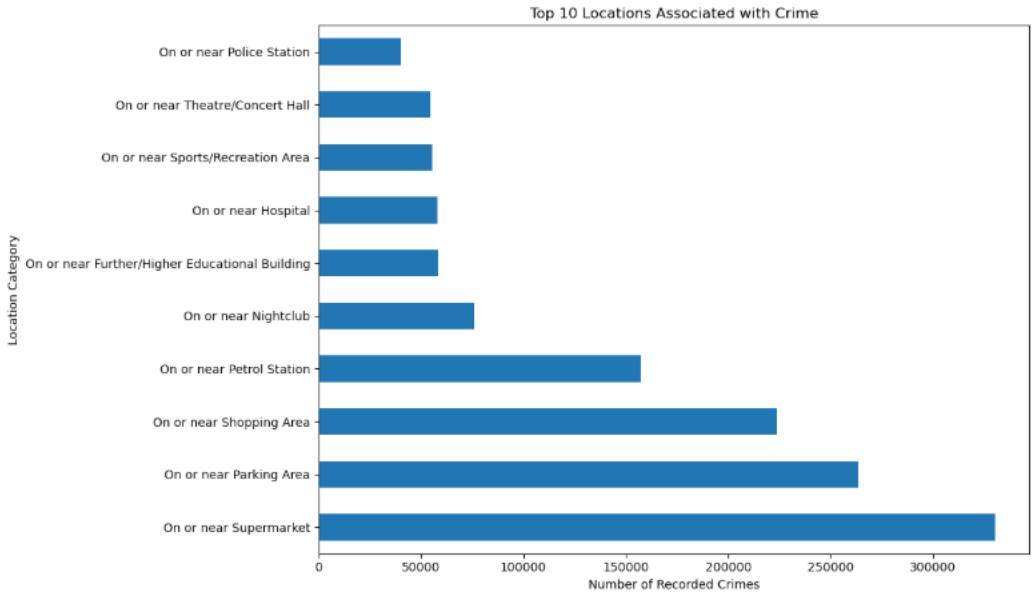


Figure 6 :Top 10 most common locations associated with recorded crime incidents.

In addition to analysing crime volumes and crime types, the dataset also contains information describing the types of locations where crimes were recorded. These location categories represent the general environment in which the incident occurred, such as commercial areas, public facilities, or transport locations.

Figure 6 presents the ten most common location categories associated with recorded crime incidents across the dataset. The results show that crimes most frequently occur on or near supermarkets, followed by parking areas and shopping areas. These locations are typically characterised by high levels of public activity and footfall, which may increase opportunities for offences such as theft, shoplifting, and anti-social behaviour.

Other commonly recorded locations include petrol stations, nightlife venues, hospitals, educational buildings, and sports or recreation areas. These environments also attract large numbers of people throughout the day, which may contribute to higher levels of recorded incidents.

Overall, the results suggest that crime incidents are often concentrated **in** busy commercial and public environments where large numbers of people gather. Understanding the types of environments where crimes occur most frequently provides additional context when interpreting crime patterns across the selected police forces.

4.5 Crime Outcomes

In addition to analysing crime volumes and types, the dataset also contains information on the outcomes of police investigations. These outcomes describe the result of a case following investigation, such as whether a suspect was charged, the investigation was completed without identifying a suspect, or prosecution was not possible.

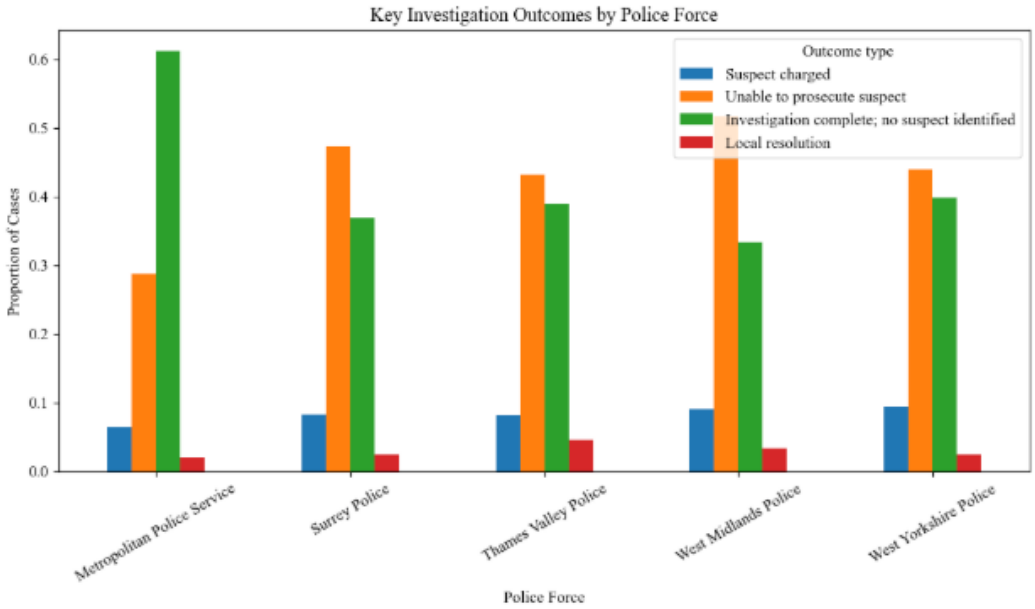


Figure 7 : Proportion of key investigation outcomes across the five selected police forces.

Figure 7 compares the proportion of several key outcome categories across the five police forces included in the analysis. The results show that “Investigation complete; no suspect identified” represents the largest share of outcomes for most police forces. This indicates that a substantial number of reported crimes are investigated but do not lead to the identification of a suspect.

The category “Unable to prosecute suspect” also accounts for a significant proportion of outcomes across all police forces. This suggests that even where a suspect may be known, prosecution may not always be possible due to insufficient evidence or other legal constraints. In contrast, outcomes such as “Suspect charged” and “Local resolution” represent a much smaller share of cases across all regions.

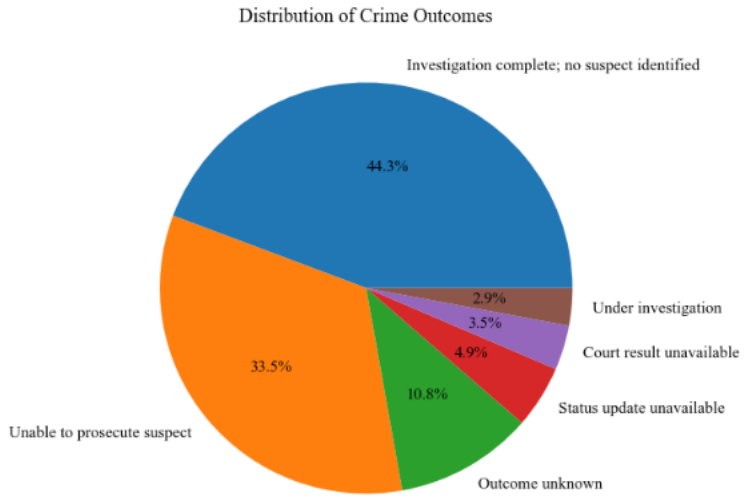


Figure 8 : Distribution of crime outcomes

Figure 8 illustrates the overall distribution of the most common outcome categories across the entire dataset. The pie chart highlights that "Investigation complete; no suspect identified" and "Unable to prosecute suspect" together account for the majority of outcomes. Other categories, such as "Outcome unknown", "Status update unavailable", and "Under investigation", represent smaller proportions of the dataset.

Overall, the outcome analysis indicates that while large numbers of crimes are recorded across the selected police forces, relatively few cases result in suspects being formally charged or locally resolved.

4.6 Key Summary Metrics

To support the comparison of crime patterns across the selected police forces, several summary metrics were calculated from the dataset. These metrics provide a high-level overview of crime volume differences between regions and help identify areas that may be more suitable for further investigation.

Table 1 summarises the relative crime levels across the five selected police forces. The table compares each police force against the Metropolitan Police Service, which records the highest crime volume within the dataset. The percentage difference highlights how much lower the total crime volume is in each region compared with London.

As shown in Table 1, Surrey Police records approximately 91.8% fewer crimes than the Metropolitan Police Service, indicating a substantially lower overall crime volume within the region. Similarly, Thames Valley Police records approximately 81.3% fewer crimes than the Metropolitan Police Service. These large differences highlight the contrast between London and the other police force areas included in the analysis.

Police Force	% Fewer Crimes vs Metropolitan Police	Average monthly crime
Metropolitan	0%	85958
West Midlands	68.3%	27212
West Yorkshire	71.1%	24869

Thames Valley	81.3%	16083
Surrey	91.8%	7044

Table 1: Summary of Crime Levels Across Selected Police Forces

The average monthly crime values also illustrate these differences clearly. The Metropolitan Police Service records an average of 85,958 crimes per month, while Surrey Police record an average of 7,044 crimes per month, and Thames Valley Police records approximately 16,083 crimes per month. These findings suggest that the overall crime environment within Surrey and Thames Valley is considerably lower than that observed within London.

Outcome Category	Percentage (%)
Investigation complete; no suspect identified	41.34
Unable to prosecute suspect	31.31
Suspect charged as part of another case	0.02
Local Resolution	2.13

Table 2:Percentage distribution of key police investigation outcomes recorded in the dataset

Table 2 presents the distribution of the most common investigation outcomes across the dataset. The results show that the most frequent outcome category is "Investigation complete; no suspect identified," which accounts for approximately 41.34% of all recorded cases. This indicates that a large proportion of reported crimes are investigated but do not lead to a suspect being identified.

The category "Unable to prosecute suspect" represents approximately 31.31% of cases, suggesting that although a suspect may be known or suspected, prosecution is not always possible due to insufficient evidence or other legal constraints.

In contrast, outcomes such as "Local resolution" represent a much smaller proportion of cases, accounting for approximately 2.13% of recorded incidents. Similarly, cases where a suspect was charged as part of another case represent a very small proportion of the dataset.

Overall, these outcome statistics indicate that while large numbers of crimes are recorded across the selected police forces, relatively few cases ultimately result in formal legal action.

4.7 Geographic Distribution of Crime

To explore the spatial distribution of crime across the selected police forces, the geographic distribution of recorded crime incidents was visualised using the latitude and longitude coordinates contained within the dataset. Figure 9 displays a sample of crime locations across the five selected police forces, with colours representing the most common crime categories.

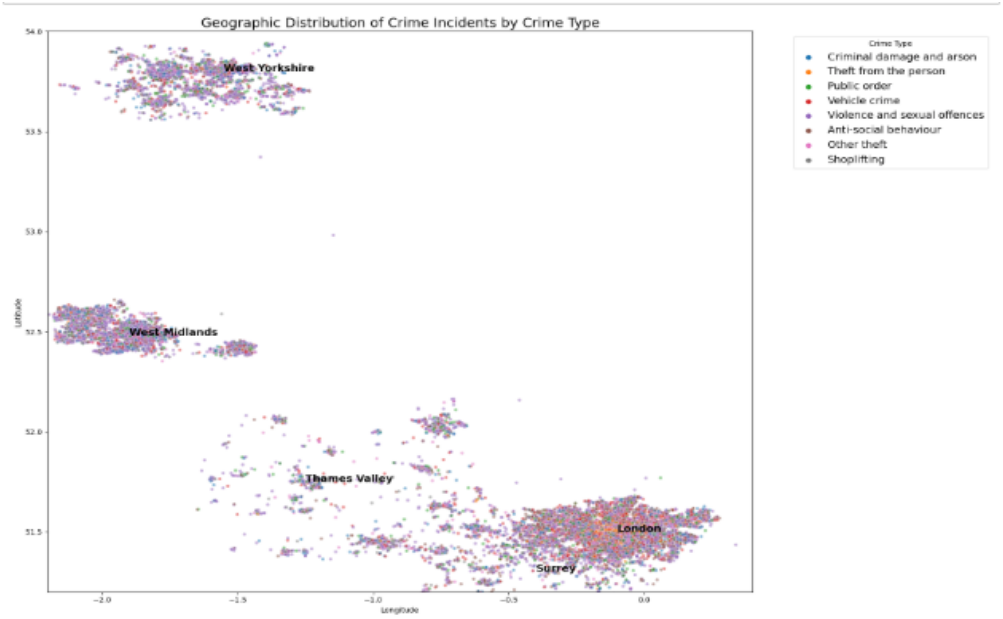


Figure 9: Geographic distribution of recorded crime incidents across the selected police forces.

The visualisation highlights clear geographic clustering of crime incidents rather than a uniform distribution across regions. The highest concentration of crime incidents is observed within the Metropolitan Police Service region, reflecting the high population density and urban nature of London. This aligns with earlier findings in the analysis, which showed that the Metropolitan Police recorded the highest overall crime volumes among the selected police forces.

Additional clusters of crime activity are visible in the West Midlands and West Yorkshire regions, corresponding to major urban centres such as Birmingham and Leeds. These regions also recorded relatively high crime volumes compared with the other police forces analysed. In contrast, crime incidents within the Thames Valley and Surrey police forces appear more dispersed and less densely concentrated. This pattern reflects the lower overall crime levels previously identified in these regions.

The geographic visualisation therefore reinforces the earlier statistical findings by illustrating how crime incidents are concentrated within major urban areas. Regions with lower crime volumes, such as Surrey and Thames Valley, show fewer and more dispersed crime locations, suggesting comparatively lower crime density within these areas.

5. Recommendations

The exploratory analysis provides a comparison of crime patterns across the five selected police forces, considering overall crime volumes, crime type distributions, trends, investigation outcomes, and common crime locations. Based on these findings, certain police forces appear more suitable for further investigation when considering regions that may be more desirable for residential living and property investment.

5.1 Recommended police forces

The evidence gathered across the exploratory analysis indicates clear differences in crime volume, crime composition, investigation outcomes, and spatial concentration across the five police forces. Based on these findings, Surrey Police and Thames Valley Police are the most suitable candidates for further investigation.

Surrey Police records substantially lower crime levels compared with the largest metropolitan police force included in the dataset. In fact, Surrey records approximately 91.8% fewer crimes than the Metropolitan Police Service, highlighting the significant difference in crime volume between these regions. This large gap indicates that Surrey operates within a comparatively lower crime environment. Surrey's proximity to London also makes it an attractive commuter region, which is relevant when considering residential property demand.

Thames Valley Police also demonstrate comparatively favourable crime levels within the dataset. The region records approximately 81.3% fewer crimes than the Metropolitan Police Service, placing it among the lower-crime environments within the five police forces analysed. Thames Valley covers key commuter counties surrounding London, including Berkshire, Buckinghamshire, and Oxfordshire, which are widely regarded as desirable residential areas. As a result, the region represents another strong candidate for deeper investigation.

Further analysis of these two police forces could examine crime patterns at the district or LSOA level to identify specific neighbourhoods with particularly favourable crime environments.

5.2 Police forces not selected for Further Analysis

The remaining police forces were not prioritised for further study due to comparatively higher crime volumes or the scale of analysis required.

The Metropolitan Police Service records the highest number of crimes by a substantial margin. While this reflects the large population and urban density of London, the scale and complexity of crime patterns within this region would require a much more detailed borough-level analysis to produce meaningful comparisons.

Similarly, West Midlands Police and West Yorkshire Police record considerably higher crime levels than the recommended regions. West Midlands Police records approximately 68.3% fewer crimes than the Metropolitan Police Service, while West Yorkshire Police records around 71.1% fewer crimes. Although these values appear lower than the Metropolitan Police Service, the overall crime volumes remain substantially higher than those observed in Surrey and Thames Valley.

For these reasons, further analysis should prioritise Surrey Police and Thames Valley Police when identifying regions that may be more desirable for residential living.

6. Limitations

This analysis is based on recorded police crime data and therefore reflects only crimes reported to and recorded by police forces. The comparison between police forces is based on absolute crime volume rather than population-adjusted crime rates, meaning that differences in regional population size should be considered when interpreting the findings. In addition, location coordinates in police datasets may be anonymised or approximate rather than exact incident points.

7. Conclusion

This report has conducted an exploratory data analysis of crime patterns across five selected UK police forces using street-level crime data from the UK Police open data portal. The analysis examined overall crime volumes, crime type distributions, monthly crime trends, investigation outcomes, and the locations where crimes most frequently occur.

The results show that crime levels vary significantly between police forces. The Metropolitan Police Service records substantially higher crime volumes than the other regions analysed, reflecting the size and density of London. In contrast, Surrey Police and Thames Valley Police record comparatively lower crime levels.

The analysis also highlights that Violence and sexual offences represent the largest category of crime across all regions, while investigation outcomes frequently indicate that cases are completed without a suspect being identified. Violence and sexual offences represent approximately 32% of all recorded crimes**,** making it the most prevalent crime category across the dataset.

In addition, crime incidents tend to occur more frequently in busy public or commercial locations, such as supermarkets, shopping areas, and parking areas.

Overall, the findings suggest that Surrey Police and Thames Valley Police may represent the most favourable regions for further investigation when considering crime levels in relation to property desirability. Future analysis could explore crime patterns at a more geographic level in order to identify specific neighbourhoods that offer particularly low crime environments. These findings provide an evidence-based starting point for a more detailed analysis of neighbourhood-level crime patterns relevant to real estate decision-making.